

Cambridge IGCSE™

MATHEMATICS

0580/41

Paper 4 (Extended)

October/November 2025

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **11** printed pages.

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Mathematics-Specific Marking Principles

- 1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.
- 2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.
- 3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.
- 4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).
- 5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.
- 6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	More information required
A0	Accuracy mark awarded zero
A1	Accuracy mark awarded one
A2	Accuracy mark awarded two
A3	Accuracy mark awarded three
B0	Independent mark awarded zero
B1	Independent mark awarded one
B2	Independent mark awarded two
B3	Independent mark awarded three
BOD	Benefit of the doubt
C	Communication mark
	Incorrect
FT	Follow through
Highlighter	Highlight a key point in the working
ISW	Ignore subsequent work
M0	Method mark awarded zero
M1	Method mark awarded one
M2	Method mark awarded two
M3	Method mark awarded three

Annotation	Meaning
	Misread
	Omission
Off-page comment	Allows comments to be entered at the bottom of the RM marking window and then displayed when the associated question item is navigated to.
On-page comment	Allows comments to be entered in speech bubbles on the candidate response.
	Premature rounding/approximation
	Special case
	Indicates that work/page has been seen
	Transcription error
	Correct
	Correct answer from incorrect working

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

- M** Method marks, awarded for a valid method applied to the problem.
- A** Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.
- B** Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more ‘method’ steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation ‘dep’ is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

- awrt answers which round to
- cao correct answer only
- dep dependent
- FT follow through after error
- isw ignore subsequent working
- nfwf not from wrong working
- oe or equivalent
- rot rounded or truncated
- SC Special Case
- soi seen or implied

Question	Answer	Marks	Partial Marks
1	rhombus	1	
2	3.5 or $3\frac{1}{2}$ or $\frac{7}{2}$	2	M1 for $11 - 4 = 2x$ or $-2x = 4 - 11$ or $\frac{11}{2} - \frac{2x}{2} = \frac{4}{2}$ or better
3	125	2	M1 for 125 or 55 correctly placed on diagram
4	[0]4 08	1	
5(a)	Triangle accurately completed with arcs at <i>C</i> .	2	B1 for accurate triangle with no/incorrect arcs or for accurate triangle with arcs but with <i>AC</i> and <i>BC</i> reversed
5(b)	80 to 85	1	FT <i>their</i> triangle if 0 scored in (a)
5(c)(i)	0.8	1	
5(c)(ii)	270	1	
6	$3g - 2g^2$ final answer	1	
7(a)	$-3 < x \leq 2$	2	B1 for $-3 < x$ or $x \leq 2$
7(b)	-1, 0, 1, 2, 3, 4	2	B1 for 5 correct (and no extras) or 6 correct with one extra or B1 for $-2 < x \leq 4$
8	1.13 or 1.125...	1	
9	34.2	2	M1 for figs 342 or for $k \times 60 \times 60$ or for $k \div 1000$
10	(4, 5)	2	B1 for each
11(a)	51	2	M1 for $60 \times \frac{100-15}{100}$ oe or B1 for 9
11(b)	68.40	2	M1 for $[] \times \frac{100-15}{100} = 58.14$ oe
12(a)	37	1	
12(b)	$F' \cap R$ oe	1	

Question	Answer	Marks	Partial Marks
13	18.6 or 18.59...	3	M2 for $\tan = \frac{12.6-1.5}{33}$ oe or M1 identifying the correct angle of elevation or for 18.6 or 18.59... seen and spoiled After 0 scored SC1 for [angle <i>ETP</i> =] 71.4 or 71.4[0...] or 71.41
14(a)	$\frac{4}{3} \times \pi \times 4^3 [= \frac{256}{3} \pi]$	1	
14(b)	47.6 nfww or 47.63 to 47.64... nfww	3	B2 for 52.4 or 52.35 to 52.37 or $\frac{50}{3} \pi$ nfww OR M2 for $\frac{8^3 - \frac{256}{3} \pi}{8^3} \times 100$ oe or M1 for $\frac{8^3 - \frac{256}{3} \pi}{8^3} [\times 100]$ oe or $\frac{\frac{256}{3} \pi}{8^3} \times 100$ oe
14(c)	2.11 or 2.107...	2	M1 for $\frac{256}{3} \pi \times 7.86$
14(d)	233 or 233.3 to 233.4	4	M1 for $\pi \times 3.1^2 \times h = \frac{256}{3} \pi$ M2dep for $2 \times \pi \times 3.1^2 + 2 \times \pi \times 3.1 \times \textit{their } h$ or M1dep for $2 \times \pi \times 3.1 \times \textit{their } h$ or M1 for $2\pi \times 3.1^2$

Question	Answer	Marks	Partial Marks
15	46	3	M2 for $[\text{angle } ATB] = 180 - 2 \times 67$ oe OR M1 for [obtuse] angle $AOB = 2 \times 67$ M1 for angle OAT or angle $OBT = 90$ OR M1 for angle $TBA = 67$ or angle $TAB = 67$ M1 for angle TDA or $TDB = 90$ where D is the intersection of OT and AB OR M1 for angle $BOT = 67$ or angle $AOT = 67$ M1 for angle $OTB = 180 - 67 - 90$ or angle $OTA = 180 - 67 - 90$
16	13.09	4	M1 for mid-values 7.5, 11 and 16 soi M1 for Σfx where x-values in correct interval (including boundaries) M1dep on second M1 for $\frac{\Sigma fx}{50}$
17	$[y =] -2x + 4$	3	M1 for gradient $= \frac{4-0}{0-2}$ oe M1 for answer $mx + 4$
18(a)	72	2	M1 for 8×8 or for $\frac{1}{2} \times 2 \times 8$ or $\frac{1}{2}(10+8) \times 8$
18(b)	13.5 or $13\frac{1}{2}$	2	B1 for 3.5 oe or M1 for $\frac{100 - \text{their } 72}{8}$ [+ 10]

Question	Answer	Marks	Partial Marks
19	29	3	B2 for 28.4[3...] OR M2 for 20×0.9^{28} or 20×0.9^{29} evaluated to at least 1 dp or for 0.9^{28} or 0.9^{29} evaluated to at least 2 dp or M1 for at least two trials of $[20 \times] 0.9^n$ so or for $1 > 20\left(1 - \frac{10}{100}\right)^n$ oe
20(a)	1.5 or $1\frac{1}{2}$ or $\frac{3}{2}$	1	
20(b)	$96 \times \left(\frac{1}{2}\right)^n$ oe final answer	2	M1 for answer $96 \times \left(\frac{1}{2}\right)^{n+k}$ oe or for $96 \times \left(\frac{1}{2}\right)^n$ oe seen
21	27.6 or 27.57 to 27.58	2	M1 for $\frac{1}{2} \times 9 \times 8 \times \sin 50$ oe
22	1.65 or 1.649[9...]	3	M2 for $\sqrt[25]{\frac{301.10}{200}}$ oe or M1 for $200 \times [\]^{25} = 301.10$
23	$[y =] \frac{9}{\sqrt{x+1}}$ oe final answer	2	M1 for $3 = \frac{k}{\sqrt{8+1}}$ oe or for $\frac{9}{\sqrt{x+1}}$ seen and spoiled

Question	Answer	Marks	Partial Marks
24(a)	$\frac{10}{x}$	1	
24(b)	<i>their</i> $\frac{10}{x} + \frac{5}{x+4} = \frac{7}{2}$ oe	M1	
	$20x + 80 + 10x = 7x^2 + 28x$ oe	M2	Strict FT for correctly clearing fractions from <i>their</i> three-term equation with two algebraic denominators in x and $x + 4$ and expanding all brackets Strict M1FT for correctly expressing their two algebraic fractions with two denominators in x and $x + 4$ as a single fraction or with a common denominator within a correct equation or for correctly clearing fractions from <i>their</i> three-term equation with two algebraic denominators in x and $x + 4$ but not all brackets expanded
	Leading to $7x^2 - 2x - 80 = 0$	A1	No errors or omissions
24(c)	$\frac{-(-2) \pm \sqrt{(-2)^2 - 4(7)(-80)}}{2(7)}$ oe	B2	or B1 for $\sqrt{(-2)^2 - 4(7)(-80)}$ oe or for $\frac{-(-2) - \sqrt{p}}{2(7)}$ oe or for $\frac{-(-2) + \sqrt{p}}{2(7)}$ oe or $\left(x - \frac{2}{14}\right)^2$
	-3.24 and 3.53	B1	
24(d)	2h 10min	3	B2 for 2.168 to 2.18 [h] or for 130.08 to 130.8 [min] or for 2hours 10.08 min to 2 hours 10.8 min OR M2 for $\frac{10}{\text{their positive } x} - \frac{5}{\text{their positive } x + 4}$ or M1 for $\frac{10}{\text{their positive } x}$ or $\frac{5}{\text{their positive } x + 4}$

Question	Answer	Marks	Partial Marks
25(a)	40	1	
25(b)	Correct column, from 2 to 5 and height 34	2	M1 for $\frac{102}{5-2}$
26	$2x^2 - 5x [= 0]$ or better or $2y^2 + 18y + 28 [= 0]$ or better	M2	M1 for $2x^2 - 3x - 7 = 2x - 7$ oe or for $y = 2\left(\frac{y+7}{2}\right)^2 - 3\left(\frac{y+7}{2}\right) - 7$
	$x = 0, y = -7$ $x = 2.5, y = -2$	B2	B1 for $x = 0, y = -7$, or for $x = 0$ and $x = 2.5$ or for $x = 2.5, y = -2$ or for $y = -2$ and $y = -7$ If M1B0 or M2B0 scored then SC1 for correct substitution seen of both of their x -values or their y -values into $y = 2x^2 - 3x - 7$ or $y = 2x - 7$
27	16.8 or 16.80 to 16.81	4	M3 for $\tan = \frac{5}{\sqrt{15^2 + 7^2}}$ oe or $\sin = \frac{5}{\sqrt{15^2 + 7^2 + 5^2}}$ oe or $\cos = \frac{\sqrt{15^2 + 7^2}}{\sqrt{15^2 + 7^2 + 5^2}}$ oe or M2 for $15^2 + 7^2$ or $15^2 + 7^2 + 5^2$ or M1 for indication of correct angle
28(a)	4	1	
28(b)	$\frac{1}{343}$	2	M1 for $[x =] f(1)$ or better
29(a)	63.0 or 63.02 to 63.03	3	M2 for $[\cos y =] \frac{10^2 + 14^2 - 13^2}{2 \times 10 \times 14}$ oe or M1 for $13^2 = 10^2 + 14^2 - 2 \times 10 \times 14 \times \cos y$ oe
29(b)	15.1 or 15.13 to 15.14	5	M2 for $[AD] = \frac{14 \sin 38}{\sin 97}$ or M1 for $\frac{AD}{\sin 38} = \frac{14}{\sin 97}$ oe M2 for $10^2 + (their AD)^2 - 2 \times 10 \times their AD \times \cos(their y + 180 - 97 - 38)$ or M1 for angle $BAD = their y + 180 - 97 - 38$ soi